Arista MG-ACC-70 Access Node Routing Protocol Flap Resiliency SOP

Node: MG-ACC-70 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_4181

Description

This Standard Operating Procedure (SOP) outlines the procedures for diagnosing and resolving a routing protocol flap failure on the Arista MG-ACC-70 access node at the MG Road Core site. The failure domain involves BGP route churn, OSPF adjacency loss, and ISIS LSP flooding, which can cause minor disruptions to network connectivity. The root cause of this failure is attributed to the interactions between the routing protocols and the hardware mechanics of the Arista EOS, specifically the CPU and memory utilization, as well as the configuration and settings of the BGP, OSPF, and ISIS protocols. The routing protocol flap can occur due to a variety of reasons, including misconfiguration, hardware issues, or software bugs, which can lead to increased CPU utilization, memory leaks, or packet drops, ultimately causing the routing protocol to flap. The diagnostic commands and resolution steps outlined in this SOP are designed to identify and address the root cause of the failure, ensuring minimal downtime and maintaining network resilience.

Pre-requisite Checks

- Verify that the MG Road Core site is accessible and remote hands are dispatched to the location.

- Check the status of the redundant links and ensure they are operational.

- Confirm that the Arista EOS software version is up-to-date and compatible with the hardware.

- Review the BGP, OSPF, and ISIS configurations to ensure they are properly set up and aligned with the network design.

Potential Root Causes

Cause	Probability
CPU utilization exceeding 80%	High
Memory leaks or fragmentation	Medium
Misconfigured BGP, OSPF, or ISIS protocols	Medium
Hardware issues such as faulty ASICs or port errors	Low

Diagnostic Commands & Expected Outputs

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Command: show system resource cpu, Expected Output: CPU utilization: 85%, Description: This command displays the current CPU utilization, which can help identify if the CPU is overloaded, causing the routing protocol flap.
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```
Command: show system resource memory, Expected Output: Memory available: 10%, Description: This command displays the current memory utilization, which can help identify if there are memory leaks or fragmentation causing the routing protocol flap.

Command: show bgp summary, Expected Output: BGP sessions: 5, routes: 1000, Description: This command displays the current BGP sessions and routes, which can help identify if there are any issues with the BGP configuration or setup.

Command: show ospf neighbor, Expected Output: OSPF neighbors: 10, adjacency status: up, Description: This command displays the current OSPF neighbors and adjacency status, which can help identify if there are any issues with the OSPF configuration or setup.

Command: show isis database, Expected Output: ISIS LSP flooding: 1000, Description: This command displays the current ISIS LSP flooding, which can help identify if there are any issues with the ISIS configuration or setup.
```

Resolution Steps

Step 1: Step: Reduce CPU utilization by disabling unnecessary processes, Command: no system resource cpu threshold 80, Description: This command disables the CPU threshold, allowing the CPU utilization to decrease and reducing the likelihood of the routing protocol flap.

Step 2: Step: Clear memory leaks or fragmentation by restarting the Arista EOS, Command: restart, Description: This command restarts the Arista EOS, clearing any memory leaks or fragmentation and resetting the routing protocols.

Step 3: Step: Review and correct BGP, OSPF, and ISIS configurations, Command: show bgp summary | show ospf neighbor | show isis database, Description: This command displays the current BGP, OSPF, and ISIS configurations, allowing the network engineer to review and correct any misconfigurations causing the routing protocol flap.

Step 4: Step: Verify that the redundant links are operational, Command: show interface status, Description: This command displays the current interface status, verifying that the redundant links are operational and the network is resilient.

Post-Check Verification

```
Command: show system resource cpu, Expected Output: CPU utilization: 20%, Description: This command displays the current CPU utilization, verifying that the CPU is no longer overloaded and the routing protocol flap has been resolved.

Command: show system resource memory, Expected Output: Memory available: 80%, Description: This command displays the current memory utilization, verifying that there are no longer any memory leaks or fragmentation causing the routing protocol flap.

Command: show bgp summary, Expected Output: BGP sessions: 5, routes: 1000, Description: This command displays the current BGP sessions and routes, verifying that the BGP configuration and setup are correct and the routing protocol flap has been resolved.

Command: show ospf neighbor, Expected Output: OSPF neighbors: 10, adjacency status: up, Description: This command displays the current OSPF neighbors and adjacency status, verifying that the OSPF configuration and setup are correct and the routing protocol flap has been resolved.

Command: show isis database, Expected Output: ISIS LSP flooding: 0, Description: This command displays the current ISIS LSP flooding, verifying that the ISIS configuration and setup are correct and the routing protocol flap has been resolved.
```

Roll-back Procedures

- Step: Re-enable CPU threshold, Command: system resource cpu threshold 80, Description: This command re-enables the CPU threshold, returning the CPU utilization to its original state.

- Step: Restore original BGP, OSPF, and ISIS configurations, Command: copy startup-config running-config, Description: This command restores the original BGP, OSPF, and ISIS configurations, reverting the network to its original state.

- Step: Restart Arista EOS, Command: restart, Description: This command restarts the Arista EOS, resetting the routing protocols and restoring the original state of the network.